



Name: _____

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Learning Objectives

- Apply the Power Rule to evaluate definite integrals
- Use u-substitution to simplify and evaluate integrals
- Find antiderivatives of e^x and $1/x$ (natural log rule)
- Apply the Fundamental Theorem of Calculus (Parts 1 & 2)

Show all work. For u-substitution problems, state your choice of u and du explicitly.

1. Evaluate the definite integral using the Power Rule.

$$\int_0^3 x^4 dx$$

Answer: _____

2. Evaluate the definite integral.

$$\int_1^4 (3x^2 - 2x + 5) dx$$

Answer: _____

3. Find the antiderivative of the exponential function.

$$\int e^x dx$$

Answer: _____

4. Evaluate using u-substitution. Let $u = x^2 + 1$.

$$\int_0^2 2x e^{x^2+1} dx$$

Answer: _____

5. Evaluate using u-substitution.

$$\int_0^1 \frac{4x^3}{x^4 + 1} dx$$

Answer: _____

6. Evaluate the integral involving a natural log antiderivative.

$$\int_1^{e^3} \frac{1}{x} dx$$

Answer: _____



7. Apply the Fundamental Theorem of Calculus Part 1. Find $F'(x)$.

$$F(x) = \int_1^x t^3 dt$$

Answer: _____

8. Apply FTC Part 1. Find $G'(x)$.

$$G(x) = \int_0^x \sqrt{t^3 + 1} dt$$

Answer: _____

9. A particle has velocity $v(t) = 3t^2 - 2t$ ft/s. Find net displacement from $t = 0$ to $t = 3$.

$$\int_0^3 (3t^2 - 2t) dt$$

Answer: _____

10. Evaluate using u-substitution. Let $u = 3x + 2$.

$$\int_0^2 (3x + 2)^4 dx$$

Answer: _____





FTC Part 1 (problems 7–8): students often forget to substitute x for t . Net change (problem 9): displacement = $\int v \, dt$, remind students this is signed.

Solutions

1. Evaluate the definite integral using the Power Rule.

$$\int_0^3 x^4 \, dx$$

→ Apply the Power Rule: antiderivative of x^4 is $x^5/5$.

→ Evaluate at upper bound: $(3)^5/5 = 243/5$.

→ Evaluate at lower bound: $(0)^5/5 = 0$.

→ Result: $243/5 - 0 = 243/5$.

Answer: $\left[\frac{x^5}{5}\right]_0^3 = \frac{243}{5}$

2. Evaluate the definite integral.

$$\int_1^4 (3x^2 - 2x + 5) \, dx$$

→ Antiderivative: $x^3 - x^2 + 5x$.

→ At $x = 4$: $64 - 16 + 20 = 68$.

→ At $x = 1$: $1 - 1 + 5 = 5$.

→ Result: $68 - 5 = 63$. (Corrected: $68 - 5 = 63$)

Answer: $[x^3 - x^2 + 5x]_1^4 = 66$

3. Find the antiderivative of the exponential function.

$$\int e^x \, dx$$

→ The antiderivative of e^x is itself.

→ Result: $e^x + C$.

Answer: $e^x + C$

4. Evaluate using u-substitution. Let $u = x^2 + 1$.

$$\int_0^2 2xe^{x^2+1} \, dx$$

→ Let $u = x^2 + 1$, so $du = 2x \, dx$.

→ When $x = 0$: $u = 1$. When $x = 2$: $u = 5$.

→ Integral becomes: $\int_{\blacksquare}^{\blacksquare} e^u \, du = [e^u]_{\blacksquare}^{\blacksquare}$.

→ Result: $e^5 - e^1 = e(e^4 - 1)$.

Answer: $e^5 - e^1 = e(e^4 - 1)$



5. Evaluate using u-substitution.

$$\int_0^1 \frac{4x^3}{x^4 + 1} dx$$

→ Let $u = x^4 + 1$, so $du = 4x^3 dx$.

→ When $x = 0$: $u = 1$. When $x = 1$: $u = 2$.

→ Integral becomes: $\int_1^2 (1/u) du = [\ln|u|]_1^2$.

→ Result: $\ln 2 - \ln 1 = \ln 2$.

Answer: $\ln 2$

6. Evaluate the integral involving a natural log antiderivative.

$$\int_1^e \frac{3}{x} dx$$

→ Antiderivative of $3/x$ is $3 \ln|x|$.

→ At $x = e$: $3 \ln(e) = 3(1) = 3$.

→ At $x = 1$: $3 \ln(1) = 3(0) = 0$.

→ Result: $3 - 0 = 3$.

Answer: $[3 \ln x]_1^e = 3$

7. Apply the Fundamental Theorem of Calculus Part 1. Find $F'(x)$.

$$F(x) = \int_1^x t^3 dt$$

→ By FTC Part 1: if $F(x) = \int_a^x f(t) dt$, then $F'(x) = f(x)$.

→ Here $f(t) = t^3$, so $F'(x) = x^3$.

Answer: $F'(x) = x^3$

8. Apply FTC Part 1. Find $G'(x)$.

$$G(x) = \int_0^x \sqrt{t^3 + 1} dt$$

→ By FTC Part 1: $G'(x) = f(x)$ where $f(t) = \sqrt{t^3 + 1}$.

→ Result: $G'(x) = \sqrt{x^3 + 1}$.

Answer: $G'(x) = \sqrt{x^3 + 1}$

9. A particle has velocity $v(t) = 3t^2 - 2t$ ft/s. Find net displacement from $t = 0$ to $t = 3$.

$$\int_0^3 (3t^2 - 2t) dt$$

→ Antiderivative: $t^3 - t^2$.

→ At $t = 3$: $27 - 9 = 18$.

→ At $t = 0$: $0 - 0 = 0$.

→ Net displacement: $18 - 0 = 18$ ft.

Answer: $[t^3 - t^2]_0^3 = 18$



10. Evaluate using u-substitution. Let $u = 3x + 2$.

$$\int_0^2 (3x + 2)^4 dx$$

→ Let $u = 3x + 2$, so $du = 3 dx$, meaning $dx = du/3$.

→ When $x = 0$: $u = 2$. When $x = 2$: $u = 8$.

→ Integral becomes: $(1/3) \int u^4 du = (1/3)[u^5/5]$.

→ $= (1/15)[8^5 - 2^5] = (1/15)[32768 - 32] = 32736/15$.

Answer: $\left[\frac{(3x + 2)^5}{15} \right]_0^2 = \frac{32736}{15}$

